

About Climate Angels

Climate Angels is a SEBI-registered Cat 1 angel investment fund that has invested in climate-tech-focussed startups. Climate Angels has invested in 18 climate tech startups in the last 2 years. The startups are operating in areas such as battery recycling, eco-friendly products, circular economy, wastewater management, EV financing, and others.

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This report is brought to you by Climate Angels and Massive Earth Fellowship as part of the Climate 101 series.

Carbon Capture, Utilization and Sequestration: On the forefront of reducing carbon emissions.

Issue statement: The Mountain of Carbon Emissions keeps piling up.

As per EDGAR (The Emissions Database for Global Atmospheric Research), the global GHG emissions in 2022 reached 53.786 GtCO₂e, increasing by 1.37% Y-o-Y. As to the contribution from the combustion of fossil fuels, it stood at 71.6% of the aggregate ≈ 38.51 GtCO₂e. The significant emitters were the usual suspects – China, the US, India, the EU27, Russia and Brazil, accounting for 61.6% of global GHG emissions.

To put this number into perspective, here's an [explainer by MIT](#) on how much a ton of CO₂ weighs (to create a visual). Imagine a cube almost as tall as an electric pole, which is 27 feet tall. So, the cube's dimensions would be 27 ft * 27 ft * 27 ft, and the volume of this cube is how much space a metric tonne of CO₂ would occupy. This cube weighs a tonne and has the same mass as a great white shark. Now, 53.786 GtCO₂e comes around to 53,786,000,000. This is the number of cubes made of GHGs humans emitted in 2022.

Impact of unabated carbon emissions

The Industrial Revolution is often seen as a pivotal period in modern history but is also responsible for the gradual upswing in the concentration of carbon emissions. The carbon concentration in the pre-industrialization period hovered somewhere over 280 ppm (parts per million), which hit a new high of 424 ppm in May 2023. Thus, with the advancement in technology, communications, and sciences has come the bane of increased carbon density in the air we breathe. Apart from this misgiving, there are several grave consequences brought on by the abundance of carbon, having severe environmental effects that are mentioned below:

- 1) **Severe storms** – As temperatures rise, so do the frequency and intensity of destructive storms such as cyclones, hurricanes, and typhoons. These storms are fuelled by warm water and cause personal and economic losses by destroying homes and communities. According to the Centre for Climate and Energy Solutions, major hurricanes have increased worldwide from the 1980s to the 2020s, while minor hurricanes have decreased.
- 2) **Rising ocean levels** – Oceans absorb excess heat from global warming, causing them to warm and expand, posing a threat to coastal and island communities.

Additionally, increased CO₂ levels result in more acidic oceans, endangering marine life ecosystems. Globally, the average sea level has risen over half a foot since 1900 and is expected to rise 1 to 2.5 feet this century.

- 3) **Frequent droughts** – Global warming directly leads to water shortages and heightened risk of impacting crop production. Droughts intensify the blowing of destructive sand and dust storms, effectively contributing to desertification, which causes a loss of land for growing food.
- 4) **Loss of Biodiversity** – Both terrestrial and aquatic species face grave threats from higher carbon concentrations, including forest fires, extreme weather, and adaptive diseases caused by climate change. Based on the UN's estimates, 4% of the mammal population will lose half their habitat if global temperatures increase by 1.5°C. In the marine ecosystem, live coral reefs have nearly halved in the past 150 years, and further warming risks destroying almost all remaining reefs.
- 5) **More Health-related Risks** – Climate change already impacts public health through air pollution, disease, extreme weather events, and mental health issues. Environmental factors cause about 13 million deaths per year.
- 6) **Food Scarcity** – The global rise in hunger and poor nutrition is one of the most significant effects of the changing climate. This can be attributed to the fact that fisheries, crops, and livestock may be grossly affected or become less productive due to the changing environment. In addition, heat abundance can lead to a decrease in water and grasslands for grazing, which can cause a drop in crop yields. The direct effect of rising CO₂ is the reduction in the concentration of proteins and essential minerals in most plant species, posing a severe threat to human health.

Countries worldwide have been grappling with these climate-related changes, though they have intensified over time due to inadequate policy and legislation and an absence of timely remedial action. However, on a brighter note, investments in sectors aimed at combating carbon emissions have seen an uptick; how further capital mobilisation turns out remains to be seen. On the innovation part, below are some of the sectors that have received market traction so far in slowing the pace of global climate change:

- **Energy Efficiency** – Better energy efficiency means using less energy to perform the same task, which can be achieved through improved technologies or processes. For instance, Kanin Energy, a Canadian startup, has created a waste heat-to-power conversion system for the iron and steel industries. The startup's Organic Rankine Cycle system is coupled with waste heat. This closed system's organic fluid heats up to drive the turbine and convert waste heat into electricity. The startup's waste heat-to-energy solution is a zero-emission system, improving gas compression and transportation sustainability.
- **Renewable Energy** – Renewable energy sources like wind, solar, hydroelectric, and geothermal don't emit CO₂ when generating electricity, thereby reducing carbon emissions. The Dutch startup KitePower has created a technology that facilitates harvesting wind energy with aerodynamic or aerostatic lift devices similar to kites. Aside from the waste reduction, up to 90% of their devices have proved twice as efficient as the turbines.

- **Sustainable Agriculture** – Sustainable farming practices like organic farming, hydroponics, and permaculture can reduce methane-heavy emissions from agriculture. They can also contribute to improved soil health and sequester carbon. Founded by three CSU microbiologists, Grocentia has developed microbial plant biostimulants and soil enhancers for agriculture. Their product enhances soil and plant health by utilising soil bacteria, helping absorb nutrients such as phosphorus, nitrogen, and potassium.
- **Waste Management** – Efficient waste management and recycling reduces the amount of waste sent to incinerators, lowering the GHG emissions during combustion. NAMO e-waste, working out of Delhi, collects disposed e-waste, treats it and recycles it into usable items. They also provide comprehensive and cost-effective solutions to their clients needing to dispose of their e-waste safely, legally and environmentally.
- **Carbon Capture, Utilization, and Sequestration (CCUS)** – This new-age technology involves capturing CO₂ emissions from sources like power plants and sequestering them underground. It can play an integral role in complementing other strategies to reduce emissions. Algiecel, a Danish startup, produces microalgae biomass, oil, and alternative sustainable ingredients used in cosmetics, feed, and food applications. Their revenue-sharing business model offers carbon capture as a service to industrial clients.

The scope of this report is to explore CCUS as a disruptive technology that can play a significant role in curtailing carbon emissions and helping sovereigns achieve their ambitious Net Zero Carbon commitments. According to the International Energy Agency (IEA), the world is at a critical point where the current solutions to reverse climate change are insufficient. Innovative technologies are necessary to slow down the effects of climate change. In the next part of the report, CCUS is explained as a technology that has the potential to capture and store carbon for a long time, making it an excellent aid to contemporary technology.

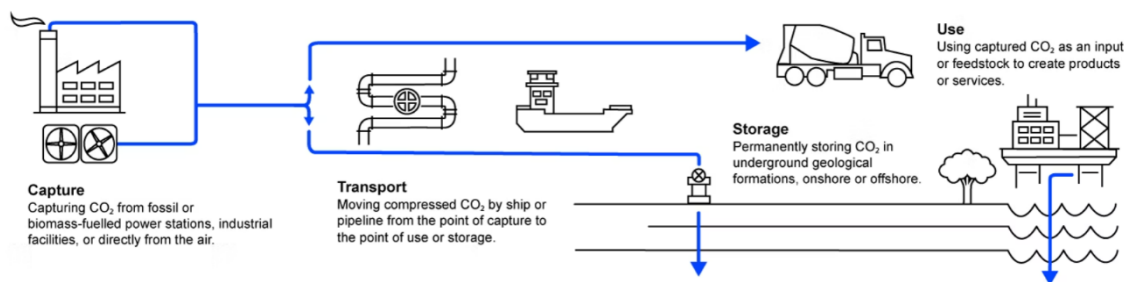
CCUS – A Groundbreaking Deep-Tech Solution

Explaining the CCUS process in detail

- 1) **Carbon Capture** – The first step in the process is to capture carbon dioxide from its sources of origin, such as industrial plants or power generation facilities that use fossil fuels or biomass as inputs. Once captured, the carbon dioxide is compressed and transported to its destination point through pipelines, ships or trucks. This destination point may be a storage facility or a place where it will be utilised.

Let's take Chart Industries as an example. This listed US company has developed Cryogenic Carbon Capture technology, which is used to reduce emissions from facilities like fossil energy power stations, cement, iron and steel and other plants. The tech has been tested for its efficiency and reduces emissions by 95 to 99% with half the cost and energy of competing processes and harmful Sox, NO_x and mercury pollutants.

- 2) **Carbon Use** – Using captured carbon on-site or as a feedstock to create products and services like concrete and plastics. A novel use of captured carbon has been as a raw material in ink production. However outlandish it might sound, GraviKy Labs has made it a reality. Their AIR-INK pen product produces high-quality, long-lasting, non-toxic pigment using carbon.
- 3) **Carbon Storage** – This step involves injecting captured carbon into deep geological formations, such as depleted oil and gas reservoirs or saline aquifers, either onshore or offshore. Carbfix, an Icelandic startup, uses this method of storing CO₂ emissions. Their process involves injecting CO₂ into basalt rocks underground, which reacts with the rock formations to form stable carbonate minerals. Once injected, the CO₂ remains stable for millennia, eliminating the need for long-term monitoring.



Source: [International Energy Agency](https://www.iea.org/carbon-capture)

There are two main categories that the developed Carbon Capture, Utilization, and Storage (CCUS) technology can be divided into:

- i. **CO₂ Capture and Utilization** Carbon capture technology involves absorbing carbon dioxide before it is released into the atmosphere. This method is widely used across various industries like cement and steel production, waste incineration, hydrogen production from fossil fuels, and power generation. The captured carbon can either be sequestered by compressing it and injecting it into porous rock layers underground, where it can remain for tens of thousands of years, or it can be used to produce various products. This process, called carbon utilisation, aims to create economic value by using carbon dioxide to make multiple products. It can be divided into 3 sub-areas:
 - **Mineralization** – Using CO₂ in concrete has the highest potential to become a large market for CO₂ in the near term. This energy-efficient process takes minimal external energy.
 - **Chemical** – Urea fertilisers and special polymers use CO₂ in small quantities. It can even be combined with hydrogen to make synthetic fuels, syngas and methanol.
 - **Biological** – CO₂ can be utilised to promote plant growth and can be captured in soils using biochar to enhance soil quality.
- ii. **CO₂ Capture, Transport, and Storage** – Carbon capture is essential for reducing the amount of carbon dioxide released into the atmosphere. Negative Emissions Technologies (NET) can achieve this, such as direct air capture with storage and

bioenergy with carbon capture and storage. These technologies remove the carbon dioxide emitted from burning fossil fuels and store it underground.

- **Direct Air Capture (DAC)** – Technology in this space collects CO₂ directly from the atmosphere, unlike carbon capture, which occurs at the point of emission origination, like at thermal plants, etc. The captured carbon can be sequestered geologically or applied to another use.
- **Bioenergy with Carbon Capture and Storage (BECCS)** – It refers to capturing and permanently storing CO₂ in the sense that biomass, which has extracted CO₂ from the atmosphere when growing, is converted to fuels or burned to generate energy. The best example is plants because they absorb CO₂ as they grow and thus remove CO₂.

The potential of NETs will have to be leveraged to remove CO₂ if net zero isn't achieved quickly enough to avoid extreme climate change.

Rising to the fore: CCUS at the centre stage.

Here are four significant ways in which CCUS can play a pivotal role in the transition of the global energy system to net zero:

1. **Tackling existing energy emissions** – The world's clean energy transition will focus on battling emissions from power stations and industrial plants. These sources of emissions produce such a large amount that other sectors will not be able to generate any emissions if we want to meet climate goals. Coal-fired power generation is particularly challenging because it contributes to almost one-third of global CO₂ emissions.

To tackle this issue, CCUS is the only alternative to repurposing existing power and industrial plants. Retrofitting these facilities with CO₂ capture devices can enable continued operation while reducing emissions.

2. **Solution for sectors in hard-to-abate emissions** – Industries like heavy manufacturing and long-distance modes of transportation such as aviation and freight rely on fossil fuels as their only energy source, as alternative sources are either too expensive or impractical. However, CCUS technology has emerged as a viable solution to remove and capture CO₂ emissions. It is one of the few options available to reduce carbon emissions in long-distance transport like aviation and shipping.
3. **Platform for low-carbon hydrogen production** – CCUS can supplement the production of clean hydrogen from natural gas or coal and offer an opportunity to bring low-carbon hydrogen into markets at lower costs.
4. **Removing carbon from the atmosphere** – There are various ways to reduce the amount of carbon in the atmosphere. These include planting trees and improving natural processes such as adding biochar to soil. In a report on limiting temperature rise to 1.5C, the IPCC highlighted the importance of carbon removal in 88 out of 90 scenarios for achieving net negative emissions. Removing carbon from the atmosphere can act as a backup plan, although not a perfect substitute, for when other technologies face challenges in being implemented commercially.

Emerging Trends in the CCUS Industry

Since CCUS itself is a recently explored segment for combating rising carbon emissions, the innovation brought about by it has nonetheless been impressive. In this portion of the report, the emerging technologies in CCUS which are in the nascent stage of development will be explained. Let's dive right in:

- 1) **Microalgae-based Carbon Capture:** Microalgae are microscopic, single-celled photosynthetic organisms found in aquatic ecosystems and play a vital role in the global carbon cycle by capturing huge amounts of CO₂ through photosynthesis. They make use of sparsely concentrated CO₂ from the atmosphere via carbon carbon-concentrating mechanism, thereby fixing CO₂ at the bottom of the seabed. As their growth is exponentially faster, microalgae can trap 10-50 times more CO₂ than terrestrial plants. Another added benefit is their capacity to deacidify seawater or wastewater used for their cultivation.

The reasoning behind using microalgae for carbon capture is that it uses CO₂ as an input for microalgae proliferation. It also releases O₂ via the process of photosynthesis. The critical things to take care of are the selection of an appropriate strain of algae and maintaining its growth conditions. Applications of this technology are abundant, like in bioenergy i.e. production of biofuels, bioproducts like animal feed and bioplastics, wastewater treatment, and many more. The residual biomass is also rich in lipids, proteins and carbohydrates, which makes them an excellent source of inputs for the production of biodiesel, biogas, biohydrogen, and bioethanol.

In summary, microalgae offer a promising solution for capturing and utilizing CO₂, contributing to climate change mitigation and sustainable practices. Their versatility and efficiency make them a powerful tool in the fight against global warming.

- 2) **Ocean-based Carbon Capture:** The approaches mentioned under this topic aim to enhance or accelerate natural biological or chemical processes of the ocean to sequester CO₂. According to an article by [Earth System Science Data](#), the ocean holds around 42 times more carbon than the atmosphere, which has resulted in ocean warming, acidification, and oxygen loss.

However, there are 2 approaches through which ocean properties can be protected and even enhanced to improve their carbon absorption abilities.

- **Abiotic Approaches** – It harnesses the physical or chemical properties of the ocean to remove CO₂ from the air. It includes solutions like electrochemical CO₂ removal which extracts CO₂ from seawater for storage on land, alkalinity enhancement involving adding certain minerals to seawater equipping more atmospheric CO₂ to dissolve into the ocean and be stored, as well as artificial downwelling.
- **Biotic Approaches** – These methods leverage the prowess of photosynthesising organisms in seawater to take up CO₂ and store that

carbon as biomass. For instance, seaweed can be cultivated and the carbon which they capture during their life will sink to the seafloor when they die. Nutrients like iron can be added to the ocean to spur phytoplankton growth, a process called ocean fertilization. Artificial upwelling aims to move deeper, nutrient-rich water to the surface, rather than adding new nutrients which takes place in ocean fertilization.

- 3) **Enhanced Mineralization:** Weathering refers to the natural process by which rocks and minerals dissolve or break down due to various factors such as water, air, acids, salts, plants, and temperature changes. During natural weathering, minerals like calcite and dolomite react with CO₂ to form stable carbonates, which are eventually deposited in soils, sediments, or oceanic environments. The process removes CO₂ from the atmosphere and provides long-term storage. Enhanced mineralization, also known as enhanced weathering, accelerates these natural processes by which various minerals absorb CO₂. Enhancing this weathering process would begin with the mining of specific kinds of rock, such as olivine or basalt, which could be done by grinding these rocks into powder and spreading it over soils, where it would react with the air to form carbonate minerals that provide long-term carbon storage.

CCUS Policy Implementation across Countries.

The Global West is leading the way in integrating the application of CCUS technology within the economy. They have established an ecosystem that incentivises initiatives driving innovation in this area. Several policy measures have been adopted to encourage market participants to mobilise capital towards developing CCUS technology, including direct capital grants, tax credit setoffs, carbon pricing schemes, subsidies, and regulatory requirements.

Here, we look at the undertaking of different nations in setting the stage to help in the proliferation of CCUS:

- 1) **United States of America:** The US is the world leader in CCUS deployment, having a total capacity to capture carbon to the tune of 25 mtpa. To show their commitment to this space, the below steps have been taken:-
- **CCUS Guidance** – The Council on Environmental Quality introduced the guidance to assist federal agencies with regulations and permit CCUS activities in the US. The guidance is intended to increase support for CCUS research, development, demonstration, and deployment, enhance Section 45Q tax incentive for CCUS, and ensure that CCUS technologies align with community perspectives.
 - **Utilizing Significant Emissions with Innovative Technology (USE IT) Act** – The USE IT Act extends the 45Q tax credit for CCUS for two years. It supports carbon utilisation and direct air capture research. The Act was introduced to advance critical research, development, and demonstration of direct air capture and carbon utilisation technologies.
 - **Section 45Q Tax Incentive** – Section 45Q of the Internal Revenue Code provides a tax credit for businesses that capture carbon emissions, as the US government wants to boost the business case for CCUS and mobilise investments in new projects.

- Storing CO₂ and Lowering Emissions (SCALE) Act – The Act aims to assist the development of CO₂ transit and storage infrastructure to support CCUS and CO₂ removal technologies. The Federal funding program aims to reduce overall costs by enabling economies of scale through interconnected transport systems.

2) European Union: The EU has developed and deployed various measures to support, fund and incentivise CCUS. These are mainly grant funds and legislative initiatives for CCUS, which have played an instrumental role in setting the platform for CCUS demonstration projects across the EU:-

- Legal and Regulatory Framework for CCUS – The International Energy Agency has published an extensive handbook for policymakers and regulators on establishing and updating legal and regulatory frameworks for CCUS. It identifies 25 priority issues that need to be addressed for an impactful deployment of CCUS.
- CCUS Roadmap to 2030 – This roadmap identifies the action necessary for large-scale development and deployment of CCS and CCU from 2020 to 2030. It is built on the work done within the CCUS-SET-Plan (Strategic Energy Technology Plan) developed by the European Commission to accelerate the development and deployment of low-carbon technologies.
- The EU Emissions Trading System (ETS) – The EU ETS is the world's first GHG emissions trading scheme applicable to all EU countries, including Iceland, Liechtenstein, and Norway. The ongoing 4th trading phase was revised to ensure emissions reductions in line with the EU's 2030 emissions reductions target, approx. 40% reduction in GHG emissions by 2030 as compared to 1990 levels. As per the latest data, participants in the EU ETS scheme reduced their GHG emissions by 35% in 2019 compared to 2005, indicating the apparent success of the scheme.
- European Green Deal – Approved and implemented in 2020, the deal aims to make the EU climate-neutral by 2050. The 2030 GHG emissions reduction target is 55% below 1990 levels, which can be attained through reviewing existing climate goals, redesigning public policies and introducing policies across energy taxation, energy efficiency, transport fuel regulations, land use, forestry and agriculture regulations, and the carbon border adjustment mechanism. Cross Border Adjustment Mechanism prevents carbon leakage and levels the playing field between goods produced in the EU and imported goods from a country with lower or no legislation regarding controlling carbon emissions.

3) Japan: The Japanese government views CCUS tech as key to achieving carbon neutrality. To further bolster their belief, they have outlined a long-term roadmap for the systematic implementation of CCS. The strategic plan brought into effect shows their commitment to investing in the development and deployment of CCS:-

- CCS Long-term Roadmap – Introduced by Japan's Ministry of Economy, Trade, and Industry in 2023, it is a tactical plan that aims to boost the deployment of CCS technology to help Japan achieve Net Zero by 2050. It is designed to consider the perceived social costs and take steps to

minimise them. The target is to reach a CO₂ storage capacity of 6–12 million tonnes of CO₂ per year by 2030 and 120–240 by 2050.

- 4) **Canada:** Canada plans to reduce its 2030 GHG emissions by 40% compared to 2005. To attain this ambitious goal, Canada intends to adopt a carbon pricing mechanism with excessive carbon emissions exceeding certain limits to be taxed based on the carbon pricing mechanism, which includes:-
- **Output-Based Pricing System (OBPS)** – The OBPS is a performance-based carbon emissions system wherein a standard level of carbon emissions is set for each industry. Facilities emitting more than the set limit shall be charged for the additional emissions on a per tonne of CO₂ basis. This performance-based system incentivises industries to keep their energy efficiency in check and limit Scope 1 emissions through CCUS measures.
- 5) **United Kingdom:** The UK aims to have CCUS projects capturing 10 mtpa CO₂ by 2030 and scale it up to 20–30 mtpa subsequently. The UK has eliminated coal-based electricity generation, increased industrial and building efficiency, and reduced methane emissions from fossil fuels. To reach net-zero emissions by 2050, the UK will need to capture 47 mtpa of CO₂ per year and to aid that; the UK government has allocated \$60 million for investment in CCUS demonstration projects.

Setting a precedent, the UK government is considering setting up dedicated projects in different parts of the CCUS value chain, including CO₂ transit and storage. This initiative will benefit the first movers ready to build CCUS infrastructure and guarantee a rate of return. Also, the focus is on creating CCUS clusters in industrial hubs around the country to capture carbon from fossil fuel power generation, biomass power production, steel production, etc. The UK government also supports the oil and gas industry in lowering carbon emissions by supporting joint investments of up to \$4 Billion in CCUS projects.

- 6) **Australia:** Australia has proactively reduced carbon and achieved net carbon neutrality by 2050. The policies through which Australia is going to execute are as follows:-
- **Low Emission Technology Statement (LETS)** – The Australian Government provides grant funding, regulation amendment and policy support for promoting the development and deployment of CCUS infrastructure. Through grants, it intends to invest up to AUD 20 billion in low emissions technology, including CCUS.
 - **Carbon Capture, Use and Storage Development Fund:** The CCUD Fund of AUD 50 million supports six projects for developing carbon capture technology and demonstration in different industries along with CO₂ storage and utilisation.
 - The Australian Government has also implemented emission reduction performance-based regulatory standards, a pre-requisite for the approval of projects. An instance is the Gorgon LNG project, where a necessary condition was the injection of at least 80% of the CO₂ emitted by the plant.

Developing a comprehensive framework for CCUS in India is underway to create viable markets for CCUS projects. The main challenge is to make it profitable enough for the private sector to consider investing in it. To make CCUS a reality in India, the following requirements must be met: provision of direct capital grants, tax credits, carbon pricing schemes, regulatory requirements, and public procurement preference for low-carbon products.

One proven method of CCUS adoption is the establishment of CCUS hubs, which are geographic concentrations of interconnected businesses, suppliers, and related organisations. In India, the development of such hubs could be a game-changer as they enable economies of scale and reduce unit costs, ultimately reducing the duplication of infrastructure planning and development phases. Additionally, shared infrastructure makes capturing CO₂ at smaller industrial facilities possible, which would be impractical and uneconomical to do with standalone CO₂ transport and storage infrastructure.

The structure of a CCUS hub is quite simple. It involves locating an industrial emitter or a group in one place to provide significant CO₂ sources. Carbon capture technology is utilised to capture CO₂ emissions from these sources. Once the CO₂ is captured, it is transported through a network of pipelines from different production sources and distributed accordingly. The final step is to store the captured CO₂ underground or utilise it for various purposes.

To explain the topic, the report delves into the workings of the Yorkshire Cluster, also called the East Coast Cluster. This cluster is a significant CCUS hub in northeast England, comprising two industrial regions – Teesside and Humber. The project's initial goal is to capture up to 10 million tonnes per annum (mtpa) of CO₂, with the capability of capturing and storing up to 27 mtpa of CO₂ by 2035, accounting for almost half of all emissions from the UK's industrial clusters. CO₂ capture sources include electricity generation, blue hydrogen, chemicals, waste processing, fuel refining, and aviation fuel. The CO₂ collected from the hub will be stored in a saline aquifer, the Endurance Reservoir, located 145 km offshore. The project has received about USD 39 million of public funding from the UK Research and Innovation Industrial Decarbonisation Challenge fund.

The participating companies can avoid the cash outflow on emissions certifications by utilising pooled transportation and storage facilities, which makes CCUS less expensive. However, to critically assess the effectiveness of CCUS clusters, it is necessary to bridge the funding gap for large-scale deployment. The Government should provide support and policies to incentivise carbon capture and sequestration to fund the capital costs, manage risks of CCUS projects, and instil confidence amongst emissions sources in the cluster to implement their plans. Additionally, combining organisational costs, getting public acceptance, pooling professional services, and CO₂ measurement and accounting services bring several benefits.

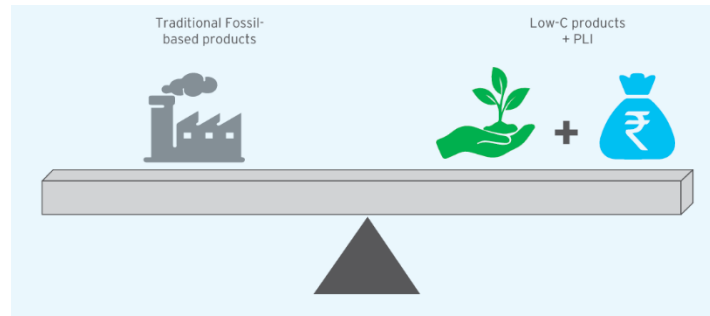
Major Risks associated with CCUS.

The key risks associated with the CCUS value chain should be addressed as it involves complex multi-stakeholder relationships that need to be considered before large-scale adoption:

- 1) **Cost of Implementation** – CCUS implementation involves significant upfront costs for building facilities, pipelines for transportation, and storage infrastructure. High capital expenditure can be a barrier to the widespread adoption of CCUS technology, especially for industries with slim profit margins.
- 2) **Storage Integrity** – Risks include the potential for leakage through geological faults or fractures, inadequate sealing of injection wells, and even CO₂ escape through porous rock formation. Monitoring and verification techniques are necessary to immediately detect and address any leakage issues.
- 3) **Environmental Impact** – Using captured CO₂ in enhanced oil recovery or manufacturing processes can have environmental consequences depending on the specific application. The capture process can have negative environmental impacts if it relies on energy sources that generate emissions.
- 4) **Public Perception and Acceptance** – Concerns about safety, environmental impact, and long-term liabilities may arise within communities near CCUS facilities or storage sites. Effective stakeholder management is paramount in building trust and assuaging public concerns.
- 5) **Regulatory and Policy Risks** – Alterations in government policies, regulations, or financial incentives can impact the viability and profitability of CCUS projects. Frequent changes in the regulatory framework also increase uncertainty and deter investment in CCUS technology.
- 6) **Technological Risks** – The CCUS technology is nascent and might face technical headwinds related to scalability, efficiency, and reliability. R&D efforts address technological risks but require targeted investments.
- 7) **Price of Green Products** – The issue of inadequate prices for green products creates roadblocks, making them uncompetitive compared to their contemporary counterparts. Feed-in tariffs have been implemented to address this risk, but only until they become cost-competitive. Ultimately, higher costs must be passed on to the customers and businesses.

CCUS Financing Mechanism and Feasibility

To ensure that low-carbon products are competitive with traditional fossil fuel-based products, it can be helpful to implement public procurement programs that favour low-carbon products or provide a guaranteed minimum price for them. This can help level the playing field for all participants. Government incentives such as tax credits, cash credits, or Production Linked Incentives (PLI) can also help accelerate the adoption of CCUS technology.



Source: [NITI Aayog](#)

The report delves into public and private funding sources available to finance CCUS projects:

- 1) **Green Bonds** – The market for Green Bonds has grown significantly, driven by the commitments of various nations and companies to achieve Net Zero Carbon. Numerous investors are participating in this market, which has led to the introduction of sovereign green bonds in many developing countries and emerging markets. Some of the capital raised from these bonds could be used to finance CCUS projects.
- 2) **Carbon Capture and Storage Fund** – The Carbon Capture and Storage Fund was established in 2009 by ADB. This fund aims to provide financing for accelerating CCS technology demonstration and deployment. Moreover, it also intends to mitigate any actual or perceived risks related to carbon capture, transport, and storage technology. All developing member countries of ADB are eligible to receive support through this fund.
- 3) **Project Finance Loans** – Project finance providers like banks and insurance corporations provide most of the capital needed for project finance. Banks include an array of commercial banks, regional banks, investment banks, as well as hedge funds.
- 4) **Development Financial Loans** – These loans are more specialised because the original lender has strict mandates to lend for sustainable development in emerging markets. Examples include the International Finance Corporation, the World Bank, the Asian Development Bank, etc.

Apart from these debt financing solutions, potential equity sources include large corporations, technology, operational investors, and sequestration site operators like crude oil and natural gas producers.

Stakeholders in the CCUS ecosystem

The interdependency among the stakeholders must be addressed to give impetus to developments in the CCUS arena. These entities collaborate to address the challenges of reducing carbon dioxide emissions and mitigating climate change:

- 1) **Government Entities and Policymakers:** National Governments set the tone for CCUS adoption by creating policies, regulations, and incentives to spur action. National climate goals influence the adoption of CCUS. Countries committed to reducing GHG emissions prioritise CCUS as part of their strategies. Funding

research and pilot projects and establishing emission reduction targets are part of their role. Global climate agreements encourage countries to work together on CCUS initiatives.

- 2) **Industry and Corporations:** Industries, specifically those with significant carbon emissions, are crucial stakeholders. They invest in CCUS projects, explore feasibility, and collaborate with research institutions to check the extent of their actions on the environment.
- 3) **Research Institutions and Academia:** Scientists, engineers, and domain experts study CCUS technologies, assess their feasibility, and improve efficiency. Private companies and startups focus on developing novel CCUS technologies. They collaborate with research institutions to refine their solutions.
- 4) **Environmental Organizations and NGOs:** Advocacy groups raise awareness about CCUS, educate the public, and advocate for adopting sustainable practices. NGOs track CCUS projects, assess their environmental impact and hold stakeholders accountable.
- 5) **Financial Institutions and Investors:** Banks and other financial institutions fund CCUS projects. Their support is crucial to providing capital for scaling up the technology and ensuring economic viability. Startups working on CCUS innovations rely on venture capital to accelerate their development.
- 6) **CCUS Hubs and Collaborative Platforms:** Such platforms bring stakeholders from different sectors together. They facilitate smooth knowledge transfer, communication, and alignment of CCUS goals. Collaborations between governments, industry, and research institutions drive CCUS projects forward.

Successful CCUS implementation requires collaboration, innovation, and long-term commitment from all these stakeholders. As technology advances rapidly and awareness grows, CCUS will play an instrumental role in achieving global climate goals.

Market Size of CCUS

According to CMI, the market size of carbon capture, utilisation, and storage in Europe and the US was estimated to be \$24.02 billion in 2023 and is poised to grow at a CAGR of 14.4% from 2023 to 32. Based on their estimates, the CCUS market will touch \$92.7 billion by 2032.

Due to the emerging nature of CCUS technology, the market size of CCUS in India isn't available publicly. Furthermore, compared to the developed countries, India lacks the formulation of policies and legislation in the CCUS space; however, as awareness increases and policies evolve, India's role in the global CCUS landscape will continue to expand.

Startups operating in the CCUS space based on location.

The US

Startup Name	Founding Year	Based out of	Description	Funding	Investors
Lanza Tech	2005	Skokie, USA	LanzaTech is a nature-based carbon refining company that transforms waste carbon into the chemical building blocks for consumer goods such as sustainable fuels, fabrics, and packaging that people use in their daily lives. The company's goal is to reduce the need for virgin fossil fuels by challenging and striving to change the way the world uses carbon.	\$869 million (Last round – Series F of \$500 million in November 2022)	ArcelorMittal, Khosla Ventures, Brookfield Renewable US
Fortera	2010	San Jose, USA	The company develops a cementitious material that provides economic savings in operating costs, while reducing carbon emissions, enabling businesses to get mandate-ready and cement producers to manufacture cement that is comparable in handling, curing, and performance to ordinary cement with less carbon dioxide emissions.	\$25.6 million (Last round – Series B of \$10.6 million in December 2011)	Khosla Ventures
Opus 12	2015	Berkeley, USA	The company is developing technology to convert captured CO2 into liquid fuels and chemicals using water and electricity as inputs for carbon capture and fuel production.	\$350,000 (Last round – Seed of undisclosed grant prize money in October 2019)	Shell, Breakout Labs, DCVD
Air Co.	2017	Brooklyn, USA	Their technology offers a solution to convert CO2 into vodka. The process involves capturing carbon dioxide from the air using an absorption-based technique. The technology then uses solar energy to power the conversion process, which breaks down the captured carbon dioxide and water using proprietary catalysts. The result is alcohol production, with oxygen being the only byproduct.	\$40 million (Last round – Series A of \$30.4 million in March 2022)	Carbon Direct Capital, Greentown Labs, Urban Future Lab
Charm Industrial	2018	San Francisco, USA	Charm Industrial offers solutions to convert biomass into pyrolysis bio-oil. The biomass is converted	\$108 million (Last round – Series B	General Catalyst, Seek

			into fast pyrolysis bio-oil, which can be injected deep underground to reduce harmful emissions. It can also be reformed to produce green hydrogen, which can be used as a fuel and industrial chemical.	of \$8 million in October 2023)	Ventures, Thrive Capital
Rubi	2020	San Francisco, USA	Offer natural textiles produced from captured carbon emissions. Enzymes convert CO2 into cellulose through a net carbon-negative and fully traceable water- and land-neutral process.	\$18.6 million (Last round – Seed of \$5.15 million in November 2023)	Talis Capital, Tin Shed Ventures, Necessary Ventures
Remora	2020	Detroit, USA	This company provides solutions for capturing carbon emissions from semi-trucks. Their device can be retrofitted onto existing trucks, sitting between the truck and its trailer and attaching to its tailpipes. The company generates revenue by selling the captured carbon dioxide to concrete manufacturers and other end-users and shares the profits with its customers.	\$5.62 million (Last round – Seed of 5.5 million in August 2021)	Union Square Ventures, Y Combinator, First Round Capital
Ebb Carbon	2021	San Carlos, USA	The company uses electrochemistry to enhance the ocean's natural ability to safely store CO2 while reducing ocean acidity, providing clients with safe, scalable, and affordable ways to remove excess CO2 from the atmosphere.	\$33.8 million (Last round – Series A of \$20 million in March 2023)	Prelude Ventures, Evok Innovations, Incite

European Union

Carbfix	2006	Reykjavik, Iceland	The company's technologies design reactors that are powered solely by renewable energy or energy-from-waste and use photons to speed up the sequestration of atmospheric and industrial CO2 emissions to create carbonates, enabling businesses to procure industrial raw materials from renewable sources.	\$117 million (Last round – Grant of \$4.52 million in August 2021)	European Union, Greentown Labs
Clime works	2009	Zurich, Switzerland	The company specialises in designing and producing devices that extract carbon	\$762 million (Last round – Series E of	Microsoft, GIC,

			dioxide from the surrounding air. They offer three different models - Demonstrator, Collector, and Capture Plants - each capable of removing a varying amount of CO2. The main target markets for these devices are F&B companies that use CO2 in food processing and greenhouse owners that need aeration. The company also aims to facilitate the creation of carbon-neutral synthetic fuels using the extracted CO2.	\$650 million in April 2022)	Partners Group
PYREG	2009	Dorth, Germany	Carbon capture and upcycling technology provider offering circular economy solutions. Recycles residual materials into high-quality biochar and renewable energy with maintenance and control systems, decentralised installation, and modular setup.	NA	German Startups Group
Skytree	2014	Amsterdam, Netherlands	Skytree has developed a technology that captures CO2 directly from the air, which can be reused in various daily products and applications. The company works on prototypes, products, and pilots for different markets.	\$6.45 million (Last round – Seed of \$6 million in June 2023)	The Yield Lab, Horticoop
UP Catalyst	2019	Tallinn, Estonia	This company provides carbon nanomaterials produced using waste biomass and existing CO2 in the atmosphere. The carbon nanomaterials they create include graphene, graphite, nanofibers, and spherical carbon. These materials have a variety of use cases, including green advanced battery technologies and biomedicine.	\$4.88 million (Last round – Seed of \$4.34 million in December 2023)	Extantia, Enterprise Estonia, SmartCap
MECAWARE	2020	Bully, France	This company provides carbon capture solutions using their MECAWARE technology, which involves CO2 capture and critical metal recycling. They also recycle technological waste from end-of-life batteries and production scraps to produce raw materials.	\$42.1 million (Last round – Series A of \$42.1 million in October 2023)	Credit Mutuel Innovation, Bpifrance, Credit Agricole

Carbon Works	2021	Libourne, France	The company's photobioreactors industrialise microalgal photosynthesis by the carbon emitted by manufacturers and convert it into biomass that could be transformed into fungicides and animal feed, enabling industries to develop safe bioproducts at a large scale and reduce their carbon footprints.	\$12.2 million (Last round – Series A of \$12.2 million in March 2022)	BNP Paribas, Bpifrance, Demeter
Greenlyte Carbon Technologies	2022	Essen, Germany	GCT is an online platform that offers solutions for capturing carbon. It uses direct air capture technology that converts the air around us into eco-friendly hydrocarbons. This platform specialises in creating and promoting a low-temperature direct air capture technology with high energy efficiency. Additionally, it produces hydrogen as a by-product by utilising thermal, low-calorific waste heat.	\$8.72 million (Last round – Seed of \$4.9 million in June 2023)	Earlybird, better ventures, HEARTFELT_

The UK

Carbon Clean	2009	London	This company offers solutions for capturing carbon emissions generated by various industries. Their platform is designed to capture and deliver industrial-quality CO2 for either reuse or sequestration. They provide a modular and semi-modular system for carbon capture and solutions for industries such as steel, cement, refineries, and energy.	\$260 million (Last round – Series C of \$190 million in February 2022)	Chevron, Samsung, Icos Capital, WAVE Equity Partners
Mission Zero	2020	London	Mission Zero is a startup that aims to transform the CO2 value chain by providing cost-effective direct air carbon capture services at scale. Their technology enables them to deliver clean and affordable CO2 to consumers, making it more accessible. By doing so, they hope to offer a more competitive cost than traditional methods.	\$5 million (Last round – Seed of \$5 million in May 2022)	Anglo American, Breakthrough Energy Ventures

Phase Biolabs	2020	Nottingham	This company specialises in biotechnology-based carbon recycling processes. Their process involves taking waste CO ₂ from different industrial processes and converting it into carbon-neutral ethanol. To achieve this, renewable energy is first converted into green hydrogen through electrolysis. This hydrogen is combined with captured CO ₂ to produce neutral chemicals and fuels.	\$125,000 (Last round – Seed of \$125,000 in January 2022)	Y Combinator, Transition
Cquestr8	2020	Durham	This company provides a carbon sequestration solution that safely locks away CO ₂ using natural processes to create stable ocean bicarbonate. They focus on decarbonising cement, lime, and steel production sectors.	\$300,000 (Last round – Seed of \$300,000 in April 2021)	Counteract
Seabound	2021	London	It offers technology to capture CO ₂ emissions from ships while docking and sell the captured CO ₂ gas for utilisation or sequestration to reduce ships' carbon footprint.	\$4.7 million (Last round – Seed of \$144,000 in November 2023)	Techstars, Lowercarbon Capital, Y Combinator
Cyano Capture	2021	Oxford	The company provides affordable, long-term carbon capture on an industrial scale by harnessing genetically modified cyanobacteria, enabling industries for rapid scale-up which current mechanical and chemical capture technologies fail to deliver.	\$1.8 million (Last round – Seed of \$1.37 million in November 2023)	Desai Ventures, Kompas
OXCCU	2021	Oxford	Provide solutions that convert CO ₂ into fuels by combining CO ₂ with H ₂ from water and renewable electricity to make fuels, chemicals, and biodegradable plastics.	\$22.7 million (Last round – Series A of \$22.7 million in June 2023)	Clean Energy Ventures, Aramco Ventures

India

String Bio	2013	Bengaluru	The company specialises in developing synthetic biotechnology solutions for recycling waste carbon. It uses metabolically engineered bacteria to produce high-value products	\$24.4 million (Last round – Series B of \$20 million)	Ankur Capital, Seventure Partners, ONGC
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			such as feedstocks and create high-performance polymers from industrial waste carbon, including methane and CO2.	in July 2022)	
Graviky Labs	2016	Bengaluru	Graviky Labs has three products in its portfolio. Firstly, there is Kaalink, a unit designed to be placed inside a car engine to capture 95% of particulate matter. Secondly, there is Air-Ink, ink made from recycled air pollution. The process of creating Air Ink carefully detoxifies heavy metals and particles. Lastly, there's Jackboy, which is a plug-and-play multi-sensor platform that connects multiple sensors to a smartphone. Users can connect phone sensors to monitor air quality, alcohol, UV index, skin moisture, and more.	\$120,000 (Last round – Seed of \$120,000 in October 2016)	Angel – Mohit Goel
Biofuels Junction	2016	Mumbai	Biofuels Junction is a prominent company in India's solid biofuels industry. Their ultimate objective is to help create a cleaner environment by replacing fossil fuels while contributing to the rural economy by boosting farmer income. Biofuels Junction produces, aggregates, and promotes the monthly sale of over 9,000 tons of biomass fuel. They also offer chain consulting services.	\$390,000 (Last round – Seed of an undisclosed amount in September 2023)	Schneider Electric, Disruptors Capital, Valud Software
CarbonC raft	2016	Hubli	The company specialises in capturing air pollution and recovering carbon from it, then fusing it with a mixture of binder and reinforcement materials to build various products for construction use, enabling construction companies to upcycle carbon products.	\$350,000 (Last round – Accelerator round of \$25,000)	Deshpande Startups, ClimateLaunchpad, Villgro
GreenJams	2019	Visakhapatnam	Manufacturing of carbon-negative building materials is the main focus of this company. Its range of materials includes concrete, hempcrete, etc.	\$369,000 (Last round – Seed of \$104,000 in September 2022)	Climate Launchpad, Villgro, InsightEdge

Rest of the World

Svante	2007	Burnaby, Canada	This company provides solutions for capturing carbon from the air. Its patented technology is capable of trapping carbon emissions from various industries, including cement, steel, ammonia, aluminium, methanol, and hydrogen. Once captured, the CO ₂ is processed, concentrated, stored, or used for industrial purposes.	\$493 million (Last round – Series E of \$15 million in March 2023)	United Airlines, Chevron, Temasek
Carbon Cure	2007	Dartmouth, Canada	The company is a developer of green concrete technologies, which involves injecting CO ₂ into the concrete during its creation. This process allows the capture of carbon dioxide from large emitters, mixed with cement and other ingredients to produce stronger concrete and sets faster. The CO ₂ gas is permanently converted into embedded solid mineral carbonates.	\$169 million (Last round – Series E of \$80 million in July 2023)	Blue Earth Capital, Breakthrough Energy, BDC, Pangaea Ventures
Carbon Engineering	2009	Squamish, Canada	Are creating technologies that capture CO ₂ from the atmosphere and use it to produce clean transportation fuels, replacing crude oil.	\$110 million (Last round – Series B of \$6.75 million in November 2022)	Air Canada, Airbus, Oxy, Chevron
Loam Bio	2019	Orange, Australia	A company specialising in microbial seed coating products has been developed to aid farming businesses. The company's products help in the restoration of carbon in the soil. The process involves coating the seeds with a microbial inoculum before sowing hemp plants. This increases soil health, nutrient-rich crops, yields, and carbon offsets.	\$120 million (Last round – Series B of \$72.6 million in February 2023)	Lowercarbon Capital, Wollemi, Grok Ventures
44.01	2020	Muscat, Oman	44.01 offers carbon capture solutions that can convert CO ₂ , captured from the atmosphere, into solid rock.	\$5 million (Last round – Seed of \$5 million in August 2021)	Apollo Projects, Systema, Breakthrough Energy

Octavia Carbon	2022	Nairobi, Kenya	This company provides technology for the direct capture of carbon. Their technology can filter CO2 directly from the air, turning it into rocks underground or used as an industrial gas to replace dirty fossil fuels.	\$100,000 (Last round – Seed of an undisclosed amount in January 2024)	Catalyst Fund, BFA, Renew Capital
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